

Literature Already Answered: The Haar Basis 1-Greedy Renorming Problem

Automatic Math Discovery Smoke Test

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Source Open Problem

The original open problem appears in Dilworth, Kutzarova, Odell, Schlumprecht, and Zsak, *Renorming spaces with greedy bases*, arXiv:1403.3777, Section 5, Problem 14. The paper asks whether, for every $1 < p < \infty$, there is an equivalent norm on $L_p[0, 1]$ for which the Haar basis is 1-greedy.

5. OPEN PROBLEMS

For bidemocratic bases we were able to achieve the best possible renorming for the democracy constant (Theorem 3). For the greedy constant Theorem 4 gets arbitrarily close, but the following remains open.

Problem 13. *Let (e_i) be a bidemocratic basis of a Banach space X . Does there exist an equivalent norm on X with respect to which (e_i) is 1-greedy?*

The following special case of interest was raised by Albiac and Wojtaszczyk.

Problem 14 ([1 Problem 6.2]). *Let $1 < p < \infty$. Does there exist an equivalent norm on $L_p[0, 1]$ with respect to which the Haar basis is 1-greedy?*

The other main problem that remains open concerns the greedy constant in the general, not necessarily bidemocratic, case.

Problem 15. *Let (e_i) be a greedy basis of a Banach space X . Does there exist for any $\varepsilon > 0$ an equivalent norm on X with respect to which the basis is $(1 + \varepsilon)$ -greedy?*

This paper gives a positive answer for a large family of bases. In terms of the behaviour of the fundamental function φ , if φ has the URP, or if, on the other extreme, $\delta(\varphi) > 0$, then the answer is ‘yes’. If the basis is bidemocratic, or, more

Status

The answer is affirmative, and this was already known to the authors of the later supporting paper. This packet is therefore a literature-status record, not an original proof and not a new literature completion by the pipeline. The later arXiv paper explicitly identifies the Albiac–Wojtaszczyk question and states that it resolves it:

F. Albiac, J. L. Ansorena, M. Berasategui, and P. M. Bernal, *Isometric Renormings for Greedy Bases in Banach Spaces, with applications to the Haar system in $L_p[0, 1]$, $1 < p < \infty$* , arXiv:2603.21733.

That paper proves that for every $1 < p < \infty$, the space $L_p[0, 1]$ can be equivalently renormed so that the L_p -normalized Haar system is isometrically greedy. In the standard terminology of greedy bases, “isometrically greedy” means greedy with constant 1. Therefore the supporting paper settles Problem 14 exactly, and explicitly knows it is settling the earlier Albiac–Wojtaszczyk problem.

Proof Chain

Let $\mathcal{H}_p$ denote the L_p -normalized Haar system in $L_p[0, 1]$.

Claim. *For every $1 < p < \infty$, $L_p[0, 1]$ admits an equivalent norm for which $\mathcal{H}_p$ is 1-greedy.*

Proof. The supporting paper proves a general renorming theorem: if a greedy basis has fundamental function equivalent to m^α , with $0 < \alpha < 1$, then the ambient Banach space admits an equivalent norm under which that basis is isometrically greedy.

For the Haar system, Temlyakov’s theorem gives that $\mathcal{H}_p$ is a greedy basis of $L_p[0, 1]$ for $1 < p < \infty$. Its fundamental function is equivalent to $m^{1/p}$. Since $1 < p < \infty$, we have $0 < 1/p < 1$. Applying the general renorming theorem with $\alpha = 1/p$ gives an equivalent norm on $L_p[0, 1]$ under which $\mathcal{H}_p$ is isometrically greedy, hence 1-greedy. $\square$

Verification Notes

No numerical computation is needed for this packet. Verification is by exact statement matching:

- The original problem asks for an equivalent norm on $L_p[0, 1]$ making the Haar basis 1-greedy for all $1 < p < \infty$.
- The supporting theorem gives an equivalent norm on $L_p[0, 1]$ making the normalized Haar system isometrically greedy for all $1 < p < \infty$.
- “Isometrically greedy” is the constant-one version of greedy, so the theorem gives precisely the requested property.

Local Artifacts

- Original paper PDF: `source_paper.pdf`.
- Supporting paper PDF: `supporting_paper_2603.21733.pdf`.
- Open-problem screenshot: `figures/open_problem.crop.png`.